

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 3



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Recap of Previous Lecture



Topic

Pipeline Processing

Topic

Pipeline Cycle Time

Topic

Speed Up

Topics to be Covered



Topic

Instruction Pipeline

Topic

Pipeline Hazards

Topic

Operand Forwarding

#Q. Consider a 6-stage pipeline with delays 2, 4, 3, 5, 3 and 4 ns. This pipeline is upgraded to a new 8-segment pipeline in which each segment delay is 2 ns.

1. How much time is saved using new pipeline over old one for 100 tasks? 311 ns
2. What is the speed up of new pipeline as compared to old pipeline for 100 tasks? 2.453

$$\frac{\text{old}}{t_p = 5 \text{ ns}}$$

$$\frac{\text{new}}{t_p = 2 \text{ ns}}$$

ans 3)

speed up of new pipeline in ideal condition

$$\frac{t_{p_{old}}}{t_{p_{new}}} = \frac{5}{2} = 2.5$$



Topic : Instruction Pipeline

If pipeline processing is implemented on Instruction Cycle



Topic : Instruction Pipeline

Assume, 5 stage pipeline

- IF: Instruction Fetch
- ID: Instruction Decode & Address Calculation
- OF: Operand Fetch
- EX: Execution
- WB: Write Back

pipeline processing $\Rightarrow$ on RISC processor

*$\Downarrow$
Reg-based based arch.*



Topic : Instruction Pipeline

total no. of inst^{ns} executed (n) = 6
no. of stages (k) = 5

Clock Cycles		→													
Instructions ↓		1	2	3	4	5	6	7	8	9	10	11	12	13	14
	I1	IF	ID	OF	EX	WB									
	I2		IF	ID	OF	EX	WB								
	I3			IF	ID	OF	EX	WB							
	I4				IF	ID	OF	EX	WB						
	I5					IF	-	-							
	I6														
	I7														
	I8								IF	ID	OF	EX	WB		
	I9									IF	ID	OF	EX	WB	

True Condition
Target instⁿ →

branch ←

stalls or bubbles
(extra cycles)

no. of cycles taken
by pipeline
 $= k + n - 1$
 $= 5 + 6 - 1$
 $= 10$

stalls
for branch instⁿ = 3

Total = 13



Topic : Instruction Pipeline

w/o any branch problem = $5 + 9 - 1 = 13$
stalls due to branch instⁿ = 3
Total = 16

Clock Cycles →		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Instructions ↓	I1	IF	ID	OF	EX	WB											
	I2		IF	ID	OF	EX	WB										
	I3			IF	ID	OF	EX	WB									
	I4				IF	ID	OF	EX	WB								
	I5					IF	—	—	IF	ID	OF	EX	WB				
	I6									IF	ID	OF	EX	WB			
	I7										IF	ID	OF	EX	WB		
	I8											IF	ID	OF	EX	WB	
	I9												IF	ID	OF	EX	WB

Annotations:

- False Condition (points to I4)
- branch (points to I5)
- stalls (points to I5)
- Target instⁿ (points to I8)



Topic : Instruction Pipeline



In cycle 5, CPU decodes instⁿ I₄ as branch instruction then I₅ is removed from pipeline and no any new instruction will be fetched until condition for branch instⁿ I₄ is evaluated.

Condition is evaluated in EX phase of branch instⁿ which is in 7th cycle for I₄. Then in cycle 8 next instⁿ to be executed will be fetched as per condition.

Case 1 :-

by default condition evaluated
in EX phase.



Case 2 :-

(given in questⁿ)
in OF phase condition
is evaluated.



Case 3 :-

given in questⁿ
fetch next instⁿ after
branch instⁿ, only when
branch instⁿ is completely
executed.



Ques) Consider a 6 stage pipeline in which branch condition is evaluated in 3rd stage of pipeline.
No. of stall cycles for each branch instⁿ = 2 ?

If condition is evaluated (Target is available) for branch instⁿ,
in i^{th} stage of pipeline, then

$$\text{no. of stall cycles due to each branch inst}^n = i - 1$$

#Q. Consider a program which contains 50 instructions I1, I2 , I3 I50. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I6 and its target is instruction I47. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is 17 ?

$$\begin{array}{rcl}
 \text{No. of inst}^{\text{ns}} \text{ executed} & = & 6 \quad (I1 \text{ to } I6) \quad 6 - 1 + 1 = 6 \\
 & & 4 \quad (I47 \text{ to } I50) \Rightarrow 50 - 47 + 1 = 4 \\
 \hline
 & & 10
 \end{array}$$

no. of cycles w/o any problem = $k+n-1 = 5+10-1 = 14$

stalls due to branch instⁿ = $4-1 = 3$

Total = 17 cycles

#Q. In the above question if the pipeline has cycle time 10ns then total time required to execute the program is 170 ns?

$$= \text{Total no. of cycles} * \text{cycle time}$$

$$= 17 * 10 \text{ ns}$$

$$= 170 \text{ ns}$$

Ques) Total time needed to execute the program in prev. questⁿ if cycle time is 10 ns and branch is not taken?

Solⁿ

no. of inst^{ns} executed = 50

w/o problem, no. of cycles = $5 + 50 - 1 = 54$

stalls for branch instⁿ = $4 - 1 = 3$

Total = 57

total time = $57 * 10 \text{ ns} = \underline{\underline{570 \text{ ns}}}$ Ans.

#Q. Consider a program which contains 200 instructions I1, I2 , I3 I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I6 and its target is instruction I16. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is 198 ?

$$\begin{aligned} \text{no. of inst}^{\text{ns}} \text{ executed} &= 6 \text{ (I1 to I6)} \\ &= 200 - 16 + 1 = 185 \text{ (I16 to I200)} \\ \hline \text{Total} &= 191 \end{aligned}$$

$$\begin{aligned} \text{w/o stalls} &= 5 + 191 - 1 = 195 \\ \text{stalls} &= 4 - 1 = 3 \\ \hline \text{Total} &= 198 \end{aligned}$$

#Q. Consider a program which contains 200 instructions I1, I2 , I3 I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I8 and its target is instruction I108. If during the execution of the program the branch is not taken then number of cycles required to execute this program in the given pipeline is ____?

Ques) A program has 200 inst^{ns} I1 to I200.

H.W.

There are 2 branch inst^{ns}

Branch inst ⁿ	Target	Taken or not
I10	I36	Taken
I54	I133	Taken

Condition of branch evaluated in 2nd stage

of pipeline -

no. of stages in pipeline is 4.

Total cycles ——— ?



Topic : Pipeline Hazards

Situations that prevent the next instruction from being executing during its designated clock cycle





Topic : Pipeline Hazards

1. Structural Hazard / Resource Conflict
2. Data Hazard / Data Dependency
3. Control Hazard / Branch Difficulty





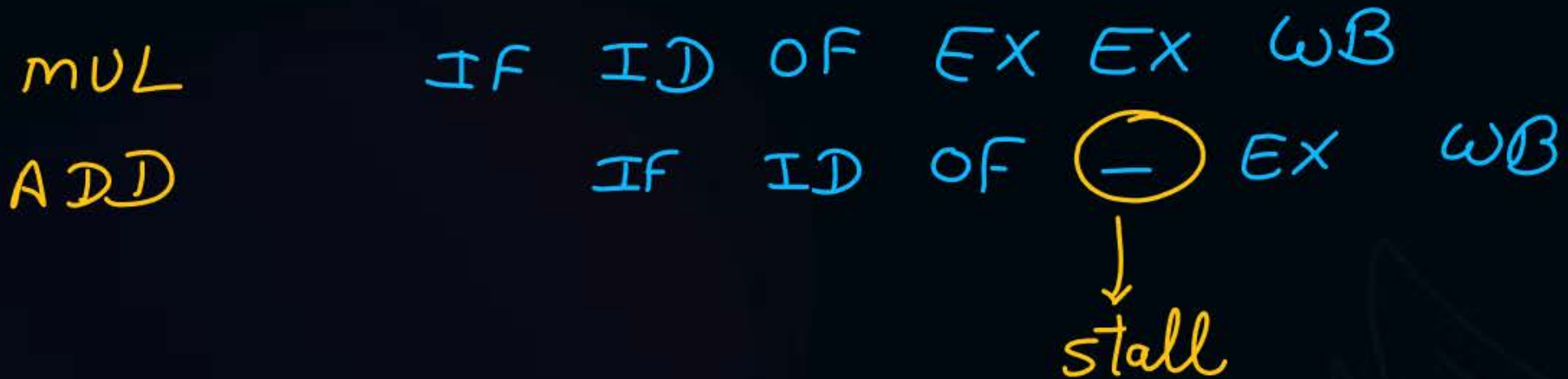
Topic : Structural Hazard

→ There is no any way (solⁿ) to save stalls here.

2 different instructions try to use same resource at same time.

each MUL instⁿ takes 2 cycles in EX phase.

ex:-



If any instⁿ takes more than 1 cycle in any stage.
↓
structural hazard



2 mins Summary



Topic

Instruction Pipeline

Topic

Pipeline Hazards

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Operand Forwarding





Happy Learning

THANK - YOU

